

Insurance Healthcare Analysis

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Introduction

The research questions:

- 1.To predict the insurance cost of patients admitted to hospitals
- 2.To determine the main factors affecting insurance costs

Methodology

1. The dataset for analysis was downloaded from Kaggle via URL1: <https://www.kaggle.com/datasets/mirichoi0218/insurance>

data is also available via github2 <https://github.com/stedy/Machine-Learning-with-R-datasets/blob/master/insurance.csv>

inspiration - <https://www.kaggle.com/code/ruslankl/health-care-cost-prediction-w-linear-regression/input3>

2. Duplicate rows were removed (one present) 4,5
3. Missing predictor variable (charges) was removed
4. Exploratory data analysis (EDA) was constructed.
5. Duplicates in the features were removed
6. Dataset was split into a 75-25 training-test set combination. Charges was log transformed since data was right skewed
7. Statistical and Machine Learning predictive models were built using the tidymodels R framework5. The models run were:
 - Linear Regression
 - Lasso Regression
 - Decision Trees
 - Random Forest
 - Gradient Boosting
 - Neural Network (MLP)

8. RMSE was used as the comparison metric $RMSE = \sqrt{\frac{1}{n} \sum_i (\hat{y}_i - y_i)^2}$

Results

Data Description and Visualisation

age: age of primary beneficiary

sex: insurance contractor gender, female, male

bmi: Body mass index, providing an understanding of body, weights that are relatively high or low relative to height, objective index of body weight (kg / m^2) using the ratio of height to weight, ideally 18.5 to 24.9

children: Number of children covered by health insurance / Number of dependents

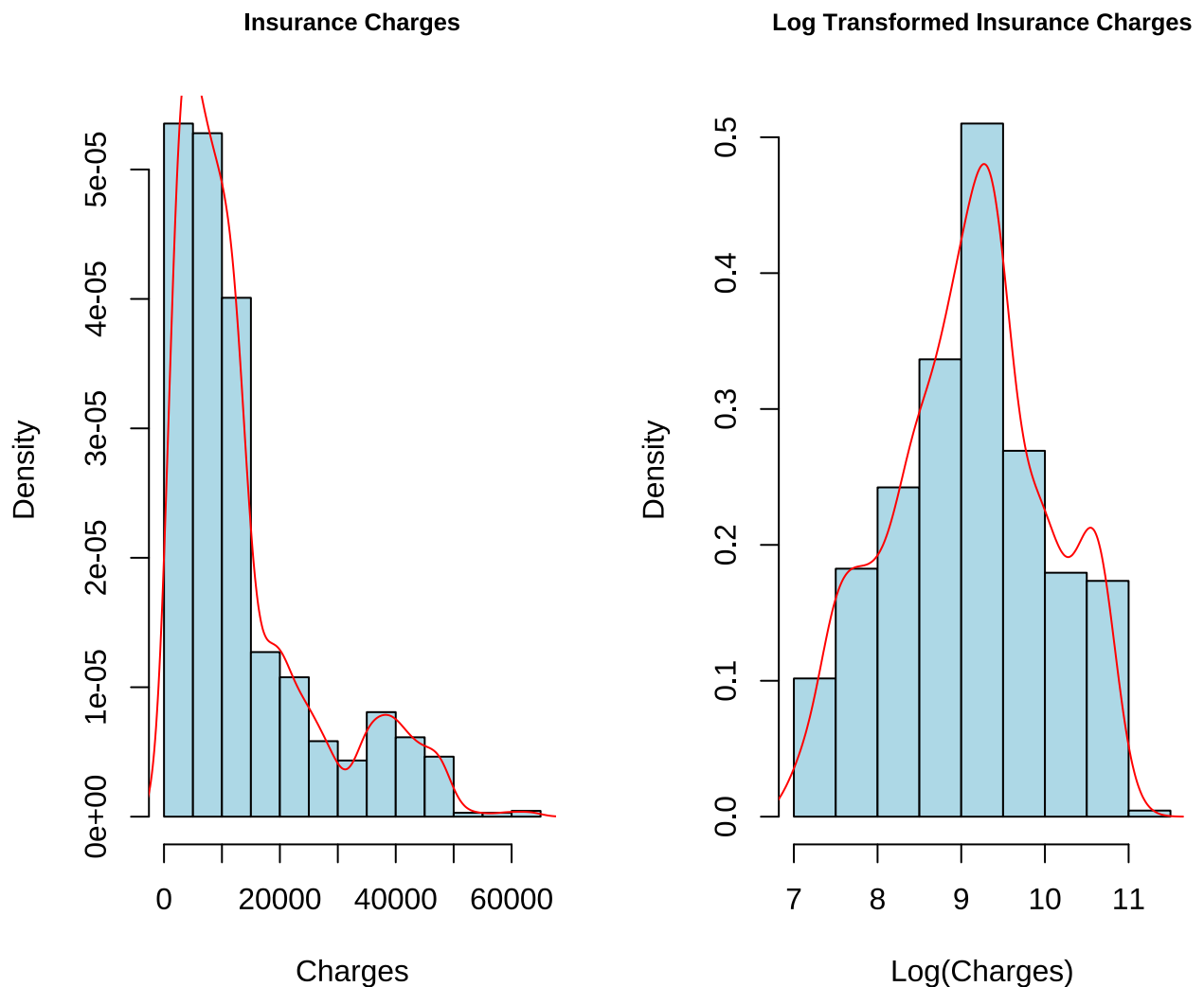
smoker: Smoking

region: the beneficiary's residential area in the US, northeast, southeast, southwest, northwest.

charges: Individual medical costs billed by health insurance

Exploratory Data Analysis

Histogram of Insurance Charges



From the graph, the charges variable has a right skewed distribution - solution is to log transform, the charges variable

Summary Statistics of dataset

Characteristic	N = 1,337 ¹
Age	39.22 (14.04)
Sex	
female	662 (49.5%)
male	675 (50.5%)
BMI	30.66 (6.10)
children	
0	573 (42.9%)
1	324 (24.2%)
2	240 (18.0%)
3	157 (11.7%)
4	25 (1.9%)
5	18 (1.3%)
Smoker	274 (20.5%)
Region	
northeast	324 (24.2%)
northwest	324 (24.2%)
southeast	364 (27.2%)
southwest	325 (24.3%)
Charges	13,279.12 (12,110.36)
Charges (Log)	9.10 (0.92)

¹Mean (SD); n (%)

Partition and Split into Training and Test

75 % of the data was used for training, while the remaining was used for the test set.

Dimensionality Reduction

Correlation Plot

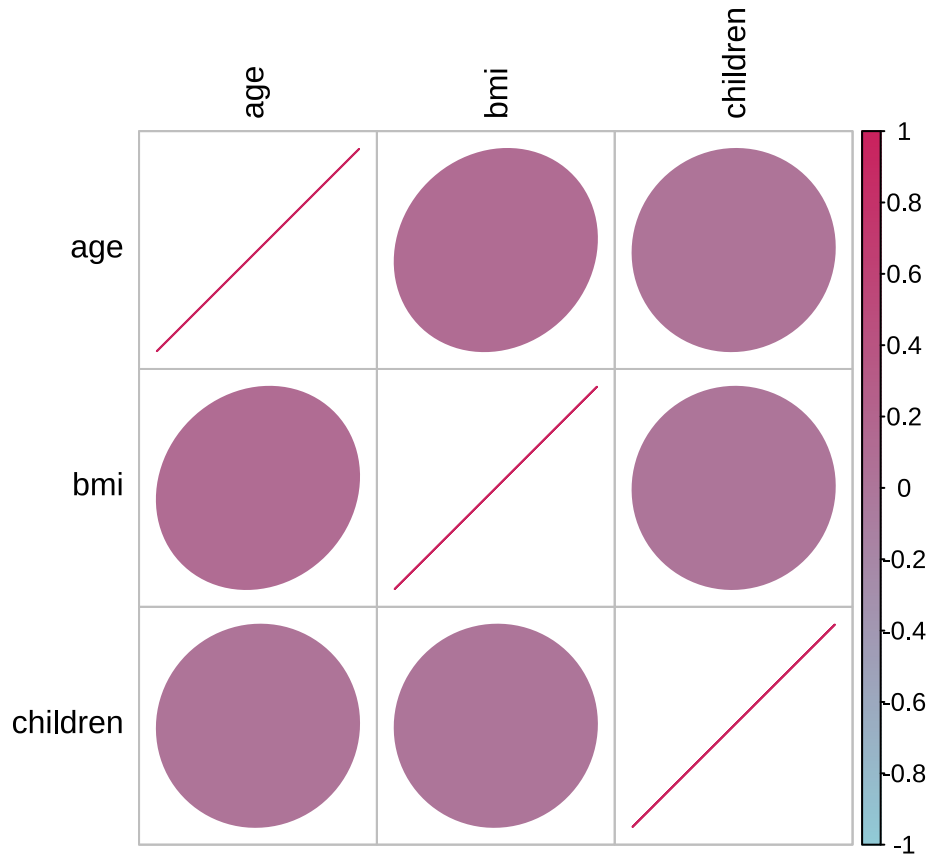


Table 2: Correlation Matrix

Variable	age	bmi	children
age	1.000	0.115	0.025
bmi	0.115	1.000	0.017
children	0.025	0.017	1.000

Models

Build Models using Tidymodels Workflow

The dependent variable (insurance cost) was log transformed for the training dataset, dummy variables were created for categorical variables (smoker, region, sex), numeric predictors were standardised by subtracting the mean and dividing by the standard deviation to ensure the same scale was used for analysis. Predictors with zero variance were removed.

To ensure speed in computing, models were run in parallel.

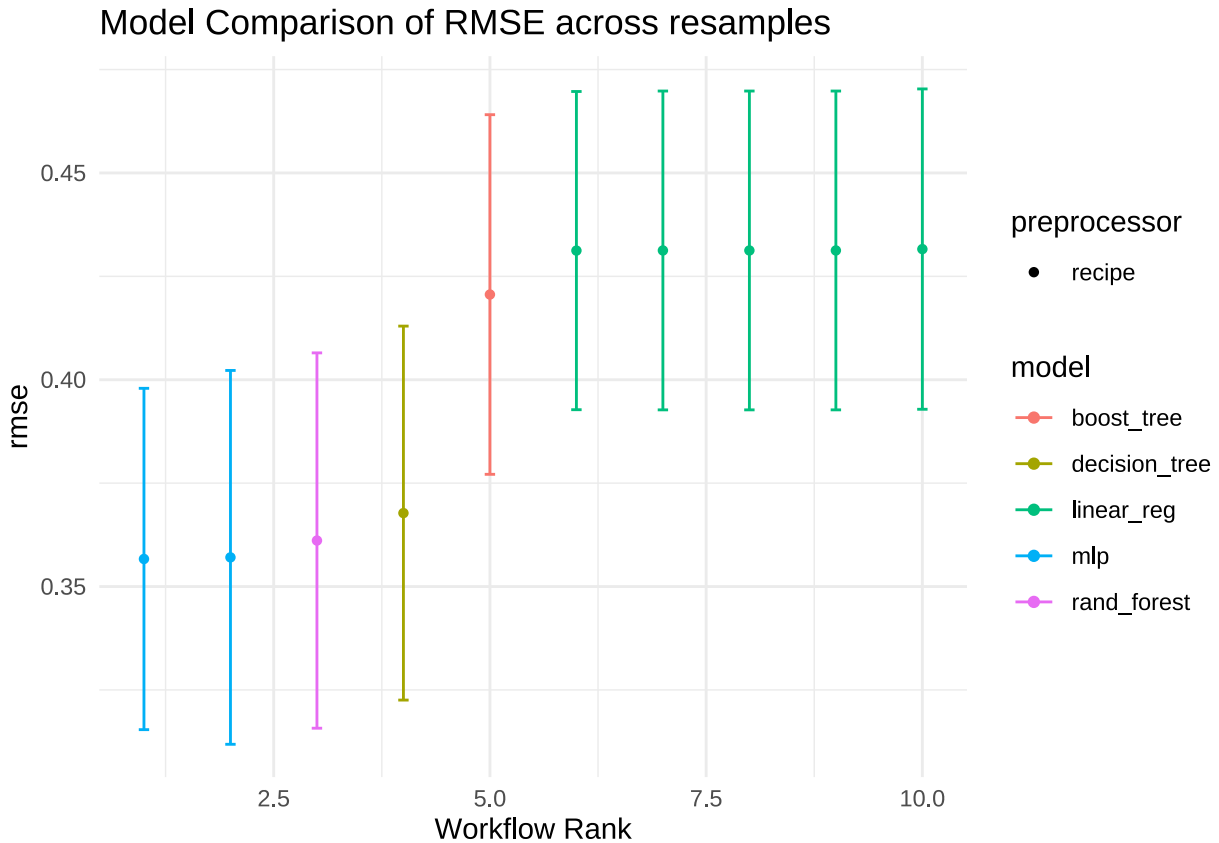
10 fold cross validation was used for tuning parameters in each model.

Table 3: Model Ranking by RMSE (Training data)

model	.config	rmse	rank
mlp	pre0_mod2_post0	0.3566590	1
mlp	pre0_mod4_post0	0.3570474	2
rand_forest	pre0_mod4_post0	0.3611171	3
decision_tree	pre0_mod2_post0	0.3677569	4
boost_tree	pre0_mod5_post0	0.4206059	5
linear_reg	pre0_mod0_post0	0.4312260	6
linear_reg	pre0_mod1_post0	0.4312580	7
linear_reg	pre0_mod2_post0	0.4312580	8
linear_reg	pre0_mod3_post0	0.4312580	9
linear_reg	pre0_mod4_post0	0.4315851	10

The best model was seen to be the neural network

Autoplot of Best Model (Smallest RSME with Training data)



The best model is the neural network. For research question 1, the neural network models appears to fit this dataset the best.

Best Results using the test data

Table 4: Best Neural Network Parameters

hidden_units	penalty	epochs	.config
3	1e-05	1000	pre0_mod2_post0

Table 5: Final Test Set Performance

.metric	.estimator	.estimate	.config
rmse	standard	0.4229218	pre0_mod0_post0
rsq	standard	0.7892791	pre0_mod0_post0

```
##
## Attaching package: 'vip'

## The following object is masked from 'package:utils':
##
##      vi
```

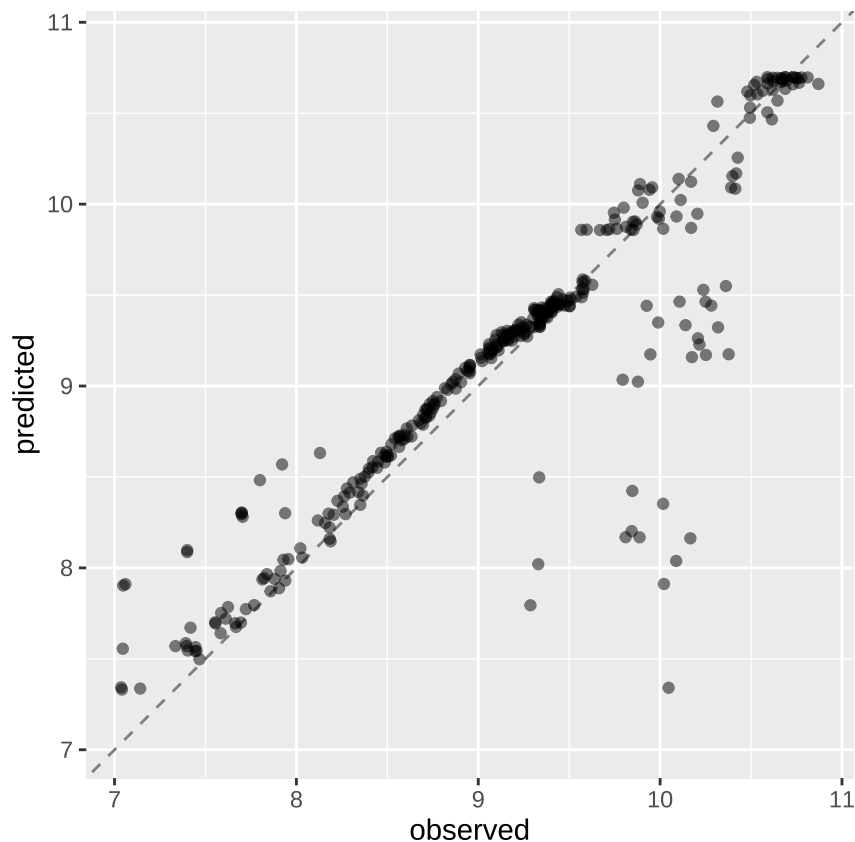
Table 6: Variable Importance for the Neural Network

Variable	Importance
smoker_yes	60.7
bmi	9.5
children	5.1
sex_male	-2.3
region_southeast	-5.1
region_northwest	-7.2
region_southwest	-11.0
age	-18.2

The RSME yields 42.3 % using the neural network model on the test set. This may indicate, there are predictors missing / unaccounted for that may explain the variation better to predict insurance costs. For example, the dataset did not include diagnosis and clinical information which may have been possible crucial predictors of insurance costs

After extracting the variable importance for each of the predictors, it appeared that the top predictors of insurance cost were smoking, bmi and children. (research question 2)

Graph showing how the model fits the test data



The predicted and observed insurance costs align mainly on the diagonal line.

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